

Assginment 1 Submission deadline: 25th Sep 2026, 11:30 pm

1. State whether each of the following observations is an example of discrete or continuous data:

1.1) **(1 point)** The number of bacterial colonies visible on an agar plate.

1.1) _____

1.2) **(1 point)** The pH of a water sample.

1.2) _____

1.3) **(1 point)** The number of mutations found in a DNA sequence.

1.3) _____

1.4) **(1 point)** The number of leaves on a seedling.

1.4) _____

1.5) **(1 point)** The elapsed time until an enzyme reaction finishes.

1.5) _____

1.6) **(1 point)** The glucose concentration in a blood sample.

1.6) _____

1.7) **(1 point)** The number of admissions to a hospital during one day.

1.7) _____

1.8) **(1 point)** The volume of culture medium in different flasks.

1.8) _____

2. **(2 points) Bird survey:** A field class records 24 birds: 5 sparrows, 4 bulbuls, 6 magpies, 3 doves, 2 kingfishers, and 4 mynas. Draw a graph that is appropriate for visualising these categorical counts.
3. **Seedling growth:** Stem length and dry mass were measured for 35 greenhouse seedlings. Length is measured in centimetres and dry mass in grams. For your convenience, you

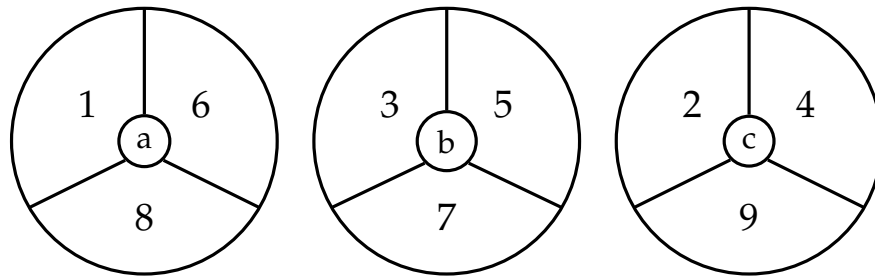
can download the source file of the table by [clicking here](#), and the file can be open by any Spreadsheet software.

ID	Length	Mass	ID	Length	Mass	ID	Length	Mass
1	2.2	0.29	13	8.4	0.62	25	16.0	0.95
2	14.5	1.27	14	12.1	0.97	26	11.2	0.69
3	16.1	1.33	15	13.4	1.03	27	15.9	1.25
4	10.4	0.70	16	15.0	1.28	28	16.3	1.25
5	18.2	1.55	17	12.7	1.04	29	19.1	1.54
6	13.3	1.13	18	16.2	1.26	30	10.3	0.88
7	1.4	0.12	19	17.1	1.27	31	15.7	1.51
8	10.1	0.91	20	4.2	0.36	32	16.0	1.15
9	15.8	1.31	21	13.1	1.06	33	9.2	0.72
10	13.0	1.09	22	3.3	0.29	34	4.5	0.66
11	15.2	1.22	23	1.8	0.37	35	0.9	0.23
12	12.3	1.07	24	13.2	1.19			

- 3.1) (2½ points) Produce a box plot of the stem lengths.
- 3.2) (2½ points) Describe the distribution of stem lengths.
- 3.3) (2½ points) Construct a scatter plot of stem length versus dry mass. Label both axes appropriately.
- 3.4) (2½ points) Does there appear to be a relationship between these two quantities? Describe its direction and strength in words.
4. (4 points) **The relation of two events:** Han Meimei spins a two-colour spinner n times ($n \geq 1$), and Li Lei spins the same spinner $n + 1$ times. Each result is either red or blue. Let $A = \{\text{Li Lei obtains more reds than Han Meimei}\}$ and $B = \{\text{Li Lei obtains more blues than Han Meimei}\}$. Are A and B mutually exclusive? Are they collectively exhaustive? Explain briefly.
5. **Weighted four-sided dice:** Li Lei has a peculiar pair of four-sided dice, and each has faces 1, 2, 3, and 4. One is in red colour, and the other is in blue colour. When he rolls the dice, the outcomes are ordered pairs (r, b) . The probability of the outcome (r, b) is proportional to $r + b + 1$.
 - 5.1) (5 points) What is the sample space for rolling the dice?

- 5.2) (7½ points) Let event $A = \{ \text{the sum of the two dice is 8} \}$, what is $\mathbb{P}(A)$?
- 5.2) _____
- 5.3) (7½ points) Let event $B = \{ \text{the sum of the two dice is odd} \}$, what is $\mathbb{P}(B)$?
- 5.3) _____
6. (10 points) **Meeting time:** Han Meimei and Li Lei plan to meet. Each arrives independently with a delay between 0 and 90 minutes, and every pair of delays in this range is equally likely. The first person to arrive waits 20 minutes and then leaves if the other has not arrived. What is the probability that they meet?
7. **A Game About Letters:** Suppose Han Meimei chooses a letter at random from the word "GENOME" and then randomly chooses a letter from the word "PROTEIN". When choosing the letter, all letters from the word are equally likely to be chosen.
- 7.1) (3 points) Write the sample space of this experiment.
- 7.2) (3 points) What is the probability of getting the outcome "EE"?
- 7.3) (4 points) Let event $A = \{ \text{At least one letter is E} \}$ and compute $\mathbb{P}(A)$.
8. **Specimen Codes:** A laboratory codes each specimen by refrigeration status (1 if refrigerated, 0 if not) and solute concentration: N (below 2 mg/mL), L (2 to below 4 mg/mL), H (4 to below 6 mg/mL), and V (6 mg/mL or above). For example, code **1L** means refrigerated with concentration from 2 to below 4 mg/mL. Consider the coding of one randomly selected specimen.
- 8.1) (2 points) Write the sample space of this experiment.
- 8.2) (2 points) Let event $A = \{ \text{Concentration is at least 4 mg/mL} \}$. Write all the outcomes in A .
- 8.3) (2 points) Let event $B = \{ \text{The specimen was not refrigerated} \}$. Write all the outcomes in B .
- 8.4) (2 points) Write all the outcomes in $B^C \cup A$.
- 8.5) (2 points) If you want to calculate $\mathbb{P}(A)$ or $\mathbb{P}(B)$, can you use the discrete uniform law? Why or why not?
9. **Transitive & Non-transitive properties:** When we talk about real numbers, we have transitive properties. For example, x, y, z are real numbers and if $x = y$ and $y = z$, then we know that $x = z$. Similarly, if $x > y$ and $y > z$, then we know that $x > z$.

However, when it comes to the “**Rock-Paper-Scissors**” game, we have non-transitive properties: rock beats scissors, scissors beat paper and paper beats rock. Now consider a new game involved in two players: Li Lei and Han Meimei. Li Lei chooses one of the three spinners pictured below, then Han Meimei chooses one of the remaining two spinners. Then they both spin their spinner, and the one that lands on the higher number is declared the winner. Assuming that each spinner is equally likely to land in any of its 3 regions.



- 9.1) **(4 points)** In terms of winning probabilities, does this game have a transitive or non-transitive property?
- 9.2) **(3 points)** If Li Lei chooses spinner @, what should Han Meimei choose to have a better chance of winning?
10. **Archery:** Let's revisit the archery example in the lecture. The target board is shown at the end of the document. It consists of 10 co-centred circles. The radius of the inner-most circle is 0.1, and the radius of the outer-most circle is 1. The circles are equally spaced, i.e. the distance between any two consecutive circles is 0.1. The score you will get when your arrow hits any particular ring is indicated below. Assume you are quite skilled, and you will always shoot on target. In addition, you can place the arrow uniformly on the target board (i.e., the probability of the arrow falling in a given region is proportional to its area). Answer the following questions:
- 10.1) **(3 points)** What is the probability of getting a score of 10 when you randomly shoot once?
- 10.2) **(3 points)** When we look at the position of the arrow, what is the sample space? Is it countable or uncountable?
- 10.3) **(3 points)** When we look at the score you get, what is the sample space? Is it countable or uncountable?
- 10.4) **(5 points)** What is the average score you will get if you shoot the target board randomly for a large number of times?

- 10.5) **(5 points)** You friend's dart is three times as likely to land in the upper half of the board as in the lower half. Across each half, the arrow falls uniformly in that region. Answer questions 10.1 and 10.2 for your friend's shooting.

